
Session 4 (2h) — Iterative Refinement & Fact-Checking

Core concepts (1h)

- **Refinement loop** — draft → critique → revise, done explicitly:
 1. Get a first draft response
 2. Ask directly: “what’s weak or missing in this?”
 3. Feed that critique back in for a revision This produces noticeably better results than hoping the first response is good enough.
- **Fact-checking generated content** — treat specific numbers, part specs, or technical claims in AI output the same way you’d treat an uncredited claim from a random forum post: fine as a starting point, not fine to put in a real report without checking it against a datasheet, your own measurement, or another source.
- **Giving corrective feedback** — instead of re-explaining the whole task from scratch, point at the specific problem (“this is too long, cut it to 3 bullet points” / “this used the wrong units, use mm not inches”) — much faster than starting a new prompt.
- **Knowing when NOT to use AI** — for anything safety-critical (wiring, voltage, mechanical load-bearing dimensions), verify against a real datasheet/reference, not just an AI explanation.

Hands-on practice (1h)

1. Pick a paragraph you’ve generated in an earlier session (docs, defect explanation, etc.).
2. Run the explicit refinement loop: ask for a critique of it, then ask for a revision based on that critique. Compare draft 1 vs. the refined version.
3. Pick one specific factual claim from any AI response you’ve gotten so far (a tolerance value, a temperature setting, an overhang angle) and verify it against an independent source (manufacturer site, datasheet, or your own test).
4. Note one example where you’d choose *not* to trust AI output outright, and why.

Checkpoint

- ☐ Ran a full draft → critique → revise loop and can point to what improved
- ☐ Independently fact-checked at least one specific AI-generated claim
- ☐ Can name a category of task where AI output needs mandatory verification